

CHUNYANG LIAO

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EMPLOYMENT

July 2023 - Present Hedrick Assistant Adjunct Professor University of California, Los Angeles

ACADEMIC TRAINING

Sept 2018 - May 2023 Ph.D. Mathematics Texas A&M University, US

RESEARCH INTERESTS

Approximation Theory, High-dimensional Probability and their applications in Scientific Machine Learning and Mathematical Data Science

PUBLICATIONS

Preprints

3. *On the Approximation of Koopman Eigenfunctions with Random Features*. Submitted.
2. *Cauchy Random Features for Operator Learning in Sobolev Space*. Submitted. With D.Needell and H.Schaeffer.
1. *Differentially private random feature model*. Submitted. With D.Needell, H.Schaeffer, and A.Xue.

Publications

8. *LR-RaNN: Lipschitz Regularized Randomized Neural Networks for System Identification*.
To appear in Proceedings of Machine Learning Research, 2025.
7. *Solving partial differential equations with random feature models*.
Communications in Nonlinear Science and Numerical Simulation, 152, 109343, 2026.
6. *Radius of information for two intersected centered hyperellipsoids and implications in Optimal Recovery from inaccurate data*.
Journal of Complexity, 83, 101841, 2024. With S. Foucart.
5. *S-procedure relaxation: a case of exactness involving Chebyshev centers*.
Explorations in the Mathematics of Data Science, Birkhäuser, 2024. With S. Foucart.
4. *On Optimal Recovery of Graph Signals*. 2023 International Conference on Sampling Theory and Applications (SampTA 2023). With S. Foucart and N. Veldt.
3. *Optimal Recovery from Inaccurate Data in Hilbert Spaces: Regularize, but what of the Parameter?*
Constructive Approximation. 57, 489–520, 2023. With S. Foucart

2. *Learning from Non-Random Data in Hilbert Spaces: An Optimal Recovery Perspective*

Sampling Theory, Signal Processing, and Data Analysis, 20, 5, 2022. With S. Foucart, S. Shahrampour, Y. Wang

1. *A Communication-Efficient Distributed Gradient Clipping Algorithm for Training Deep Neural Networks*

In Advances in Neural Information Processing Systems 35, 2022 (NeurIPS 2022). With Mingrui Liu, Zhenxun Zhuang, and Yunwen Lei.

TEACHING EXPERIENCE

UCLA (2023-):

- Instructor, Python with Application I, (Fall 2023, Winter 2025, Summer 2025, Fall 2025)
- Instructor, Python with Application II, (Winter 2024, Spring 2024, Fall 2024, Spring 2025)
- Instructor, Machine Learning, (Summer 2024)

Texas A&M University (2018 - 2023):

- Instructor, Applied analysis Qualifying Exam Preparation course (Summer 2022)
- Instructor, Mathematics for Business and Social Sciences (Fall 2021)

MENTORING

Undergraduate Students:

- Minghao Wang (Fall 2025 -)

Summer REU:

- Academic Mentor, Research in Industrial Project for Students (RIPS) Program 2024 at IPAM

Title: System Identification using Vector Fitting and Neural Networks.

Sponsors: Analog Device Inc.

The work results in a poster presentation at 2025 Joint Mathematics Meeting and a student poster presentation at AAAI student poster program.

- Research in Industrial Project for Students (RIPS) Program 2025 at IPAM.

Title: Leveraging Large Language Models (LLMs) to Enhance Survey Data. *Sponsors:* RAND Corporation

ORAL PRESENTATIONS

Invited Workshop and Conference Presentations

- *Solving partial differential equations with random features.* AMS Special Session on Randomized Methods for Data Analysis. 2026 Joint Mathematics Meeting. Washington D.C., January 4-7, 2026.
- *Cauchy Random Features for Operator Learning in Sobolev Space.* SIAM Minisymposium on Approximation Theory. 2026 Joint Mathematics Meeting. Washington D.C., January 4-7, 2026.

- *Cauchy Random Features for Operator Learning in Sobolev Space*. The 10th Annual Meeting of SIAM Central States Section, University of Arkansas, Fayetteville, Arkansas, October 11-12, 2025
- *Solving partial differential equations with random features*. The 8th Annual Meeting of the SIAM Conference Texas-Louisiana Section, UT Austin, Texas, Sept 26-28, 2025
- *Radius of information for two intersected centered hyperellipsoids and implications in Optimal Recovery from inaccurate data*. SIAM CSE 2025, Fort Worth, March 3-8, 2025.
- *Cauchy Random Features for Operator Learning in Sobolev Space*. 7th Annual Meeting of the SIAM Conference Texas-Louisiana Section, Baylor University, Waco, Texas, October 11-13, 2024
- *Optimal Recovery from inaccurate observations*. Special session 'Sparse signal learning and its applications in data science', 13th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Wilmington, North Carolina, May 31-June 4, 2023
- *Optimal Recovery from Inaccurate Data in Hilbert Spaces: Regularize, but what of the Parameter?* Minisymposium 'Data Reduction, Approximation, and Computation', International Conference on Approximation Theory and Beyond, Vanderbilt University, Nashville, Tennessee, May 15-18, 2023
- *Optimal Recovery from Inaccurate Data in Hilbert Spaces: Regularize, but what of the Parameter?* Minisymposium 'Recent Advances in Learning', 5th annual meeting of the SIAM TX-LA Section, Houston, Texas, Nov 4-6, 2022.
- *Optimal Recovery in Hilbert spaces from observational data*. 2022 Summer Informal Regional Functional Analysis Seminar (SUMIRFAS), Texas A&M University, College Station, July 29-31, 2022.

Contributed Conference Presentations

- *Cauchy Random Features for Operator Learning in Sobolev Space*. Southern California Applied Mathematics Symposium (SOCAMS), UC Riverside, April 26, 2025.
- *On the optimal recovery of graph signals*. Contributed Talks, Sampling Theory and Applications Conference (SampTA 2023), New Haven, Connecticut, July 10-14, 2023
- *On the optimal recovery of graph signals*. Contributed Talks, Inaugural CAMDA Conference, College Station, Texas, May 22-25, 2023
- *Optimal Recovery from Inaccurate Data in Hilbert Spaces: Regularize, but what of the Parameter?*, SIAM Annual Meeting, Pittsburgh, Pennsylvania, July 11-15, 2022.

Seminars

- *Radius of information for two intersected centered hyperellipsoids and implications in Optimal Recovery from inaccurate data*. Applied Math Seminar, University of Arkansas, April 17, 2024.
- *Radius of information for two intersected centered hyperellipsoids and implications in Optimal Recovery from inaccurate data*. Applied Math Seminar, UC Irvine, March 4, 2024.
- *Learning Theory from the View of Optimal Recovery*. Level Set Meeting, UCLA, May 1, 2023.

- *Optimal Recovery in Hilbert spaces from observational data.* Applied Math Seminar, University of Georgia, Athens, Sept 13, 2022.
- *Optimal Recovery in Hilbert Spaces from Exact or Inaccurate data.* Center of Approximation and Mathematical Data Analytics (CAMDA) seminar, Texas A&M university, College Station, Feb 23, 2022

PROFESSIONAL SERVICES

Reviewing: Sampling Theory, Signal Processing, and Data Analysis (2022), Sampling Theory and Applications Conference (2025), The Annals of Applied Probability (2025), SIAM Journal of Numerical Analysis (2025), Journal of Applied Analysis and Computation (2025), Chaos: An Interdisciplinary Journal of Nonlinear Science (2025), Applied Science (2025).

Journal Editorial Board:

- Mathematics, Guest Editor for the special issue “Advanced Mathematical and Statistical Methods for Modern Data Science and Scientific Machine Learning”, 2025–2026.

Organization

- Co-organizer of the minisymposium *Approximation Theory and its Applications in Scientific Machine Learning*
The 10th Annual Meeting of SIAM Central States Section, University of Arkansas, Oct 11-12, 2025